SITHUM MANUSHA

QUALITY ASSURANCE (QA) INTERN

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ABOUT ME

A third-year Information Technology & Management undergraduate at the University of Moratuwa with a solid foundation in automated testing, REST API validation, and full-stack software quality assurance. Experienced in testing complex web systems using JUnit 5, Pytest, Playwright, Vitest, and Postman, alongside real-time data synchronization verification. Seeking a Quality Assurance (QA) Internship to design rigorous test suites, validate mission-critical workflows, and ensure high-standard software reliability.

EDUCATION

University of Moratuwa (UOM) - Moratuwa, Sri Lanka **2024 - Present**
B.Sc. (Hons) in Information Technology & Management **(3rd Year | CGPA: 3.23 / 4.0)**

Open University of Sri Lanka (OUSL) - Nawala, Sri Lanka **2025 - Present**
Bachelor of Laws Honours Degree Programme (LL.B.)

St. Joseph's Vaz College - Wennappuwa, Sri Lanka **2009 - 2022**
G.C.E. Advanced Level (Commerce Stream: 2 A's & 1 B) | G.C.E. Ordinary Level (8 A's & 1 B)

FEATURED PROJECTS

1. CareerFlow — Distributed ATS Resume Optimizer & AI Interview Platform (Individual Project) **2026**

- Architected a decoupled microservices platform uniting a **Java 21 Spring Boot 3** core gateway, a **Python 3.12 FastAPI** NLP worker, and a **Next.js 14** web client deployed on Render Cloud.
- Built an automated ATS resume scoring engine using **TF-IDF vectorization & Cosine Similarity** with missing keyword classification, format health audits, and Google XYZ bullet point rewriting.
- Implemented an interactive 5-round AI mock interview simulator with manager intent breakdowns, token-bucket rate limiting (10 req/min), and achieved **comprehensive test coverage across 26 automated tests (JUnit 5, Pytest, Playwright)**.
- **Technologies:** Java 21, Spring Boot 3, Python 3.12, FastAPI, Next.js 14, PostgreSQL, Scikit-learn, Docker, JUnit 5, Pytest, Playwright
- **Live Demo:** careerflow-pink.vercel.app | **GitHub:** github.com/SithumManusha/careerflow

2. Super POS — Multi-Branch Enterprise Offline-First Retail System (Group Project) **2026**

- Built a client-side offline sync queue (**useOfflineSync** with **IndexedDB offlineQueue**) and batch retry pipelines (**/sync/push, /sync/retry, /sync/batch-retry**) to buffer offline sales, stock updates, and price changes with auto-reconnection sync to **PostgreSQL**.
- Implemented **Last-Write-Wins (LWW)** conflict detection with an interactive 3-way resolver (**Trust Local, Trust Cloud, and Smart Merge**) and **Recharts** analytics dashboards tracking P50/P95/P99 latency and sync throughput.
- Developed a live regional node topology monitor and fleet broadcast system via **Socket.io WebSockets**, alongside remote device access revocation, entity sync policies, and an audit trail with **PDF health report exports**.
- **Technologies:** Next.js, React, TypeScript, Tailwind CSS, IndexedDB, Recharts, NestJS, Node.js, Prisma ORM, Socket.io, Zod, PostgreSQL (Supabase), Turborepo
- **Live Demo:** pos-system-pos-web-app.vercel.app | **GitHub:** github.com/Ryzera/Ryzera-POS

3. Ledgerly — Autonomous AI Financial Intelligence & Collaborative Ledger (Individual Project) **2026**

- Engineered an **Autonomous AI Copilot** to prevent budget overruns via 0–100 composite health scoring, spending velocity anomaly alerts (>25% surge), and grounded natural language financial advice.
- Built an interactive **What-If runway simulator** to forecast capital purchase impacts via dynamic stress-testing sliders and 6-month Recharts curves, paired with a **real-time Server-Sent Events (SSE)** alert bell.
- Solved multi-party expense disputes by engineering **4 debt-minimization split models** and streaming PDFKit statements, backed by a type-safe RPC stack (**React 19, TypeScript, tRPC 11, PostgreSQL**) and **68/68 automated tests (Vitest)**.
- **Technologies:** React 19, TypeScript, tRPC 11, Node.js, Express, PostgreSQL, Drizzle ORM, SSE, Recharts, Vitest, Render
- **Live Demo:** ledgerly-nine-rust.vercel.app | **GitHub:** github.com/SithumManusha/Ledgerly

4. Automated Robotic Sorting System — Microcontroller Conveyor Platform (Group Project) **2025**

Level 1 Hardware Project (2024 – 2025)

- Constructed an automated conveyor sorting prototype equipped with a 4-DOF robotic arm, ultrasonic sensors, an RFID reader, and 3 sorting compartments.
- Engineered a two-finger mechanical gripper driven by an Arduino and stepper motor, calibrating motor step counts to achieve 98% pick-and-place precision across 3 RFID object categories.
- Tested sensor signal timings and synchronized robotic arm kinematics with continuous conveyor belt transport.
- **Technologies:** Arduino Mega/Uno (C++), Stepper Motors (NEMA 17, 28BYJ-48), ULN2003, Servo Motors, Integration Testing
- **GitHub:** github.com/SithumManusha/robotic-sorting-system

TECHNICAL SKILLS

- **Testing & Quality:** Automated Unit Testing, API Testing, E2E Testing, JUnit 5, Pytest, Playwright, Vitest, Postman, Jira
- **Programming Languages:** Java, Python, JavaScript, TypeScript, C, SQL
- **Web Technologies & Frameworks:** React, Next.js 14/16, Spring Boot 3, FastAPI, Node.js, Express.js, NestJS, Tailwind CSS
- **Databases & ORM:** PostgreSQL, MySQL, Redis, Spring Data JPA, Hibernate, Prisma ORM
- **Tools & DevOps:** Docker, Git, GitHub Actions (CI/CD), Swagger / OpenAPI, Render Cloud, VS Code, IntelliJ

INTERPERSONAL SKILLS

Problem Solving, Teamwork, Clear Communication, Time Management, Attention to Detail, Adaptability

CERTIFICATIONS & RELEVANT COURSEWORK

- Web Design for Beginners – University of Moratuwa (CODL) | Python for Beginners – UOM (CODL)
- AWS Cloud Practitioner Essentials – Amazon Web Services (AWS) | Introduction to Cybersecurity – Cisco

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- Member of the Organizing Committee, Official Career Fair conducted by University of Moratuwa (2026)
- Member of the Organizing Committee, Project "Rest with Hope" conducted by Leo Club, UoM (2025)
- Active Member – FIT Moments (Faculty Media Body) | Leo Club of UoM | Legal Youth Community Sri Lanka

LANGUAGES

- English: Professional Working Proficiency (Oral & Written) | Sinhala: Native Proficiency

REFERENCES

Dr. A.L.A.R.R. Thanuja (B.Sc. (Sri Lanka), Ph.D. (USA)),
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